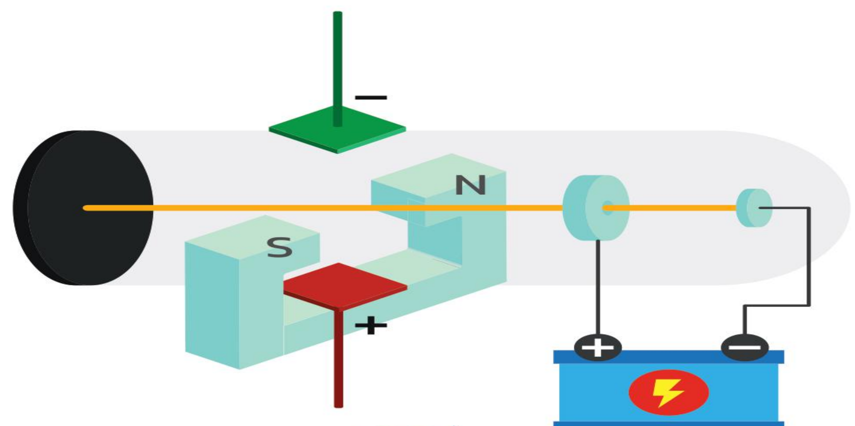
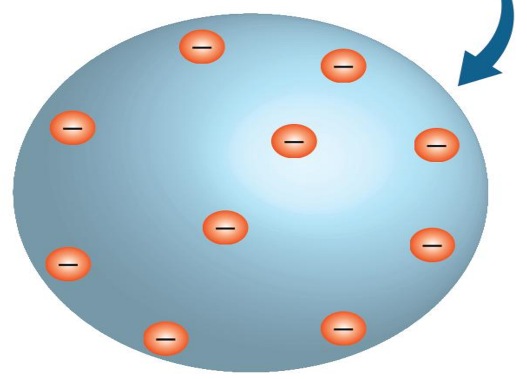
ATOMS



THOMSON'S ATOMIC MODEL

- Also known as pudding model
- Positive charge are uniformly distributed in the atom.
- Negative charge are embedded like seeds in watermelon.
- Overall atom is neutral



LIMITATIONS

- This model does not explain the presence of nucleus in the atom.
- This is not able to explain scattering of α - particles
- This is not able to explain the spectrum of atoms.

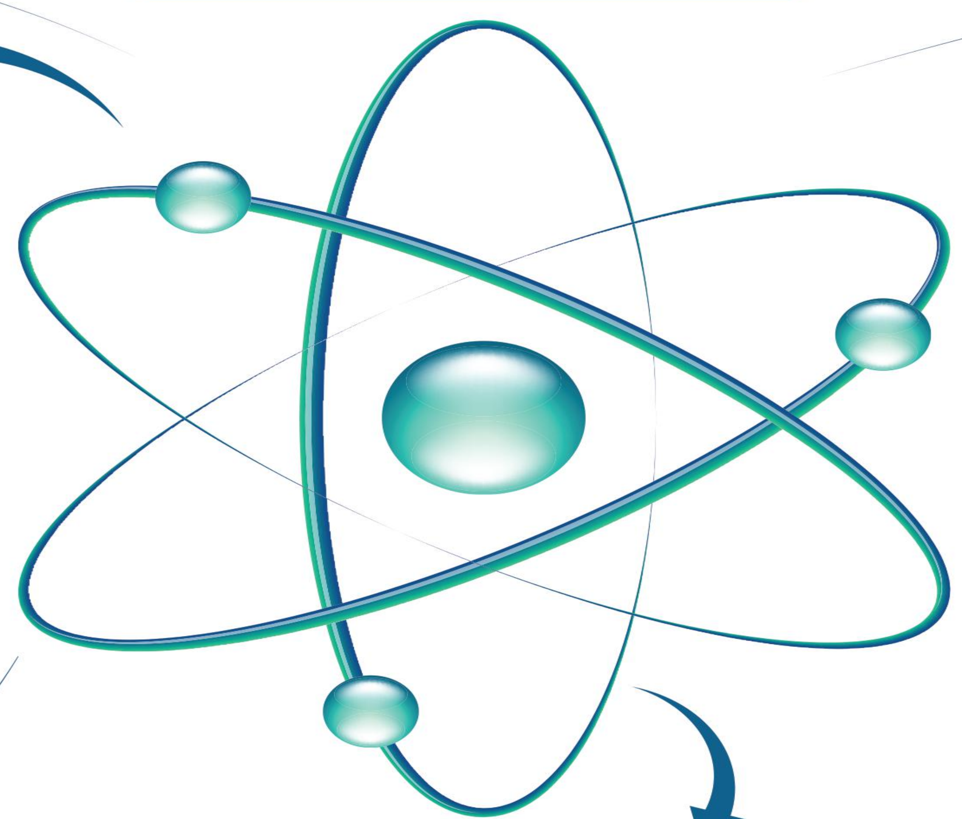
DISTANCE OF CLOSEST APPROACH

At closest approach. System only have electric potential energy.

$$K = U = \frac{1}{4\pi\varepsilon_o}\frac{(2e)(Ze)}{d}$$

$$\Rightarrow d = \frac{1}{4\pi\varepsilon_o}\frac{2ze^2}{K}$$

$$\Rightarrow d = \frac{1}{4\pi\varepsilon_o}\frac{2ze^2}{(\frac{1}{2}mv^2)}$$



BOHR'S MODEL

(i) Valid for only one – electron atom.
(ii) Electron is revolving around the nucleus in a stable orbit.
(iii) Attractive Coulomb force between electron and nucleus is equal to the centripetal force of electron

$$\frac{Ze^2}{4\pi\varepsilon_o r} = \frac{mv^2}{r}$$

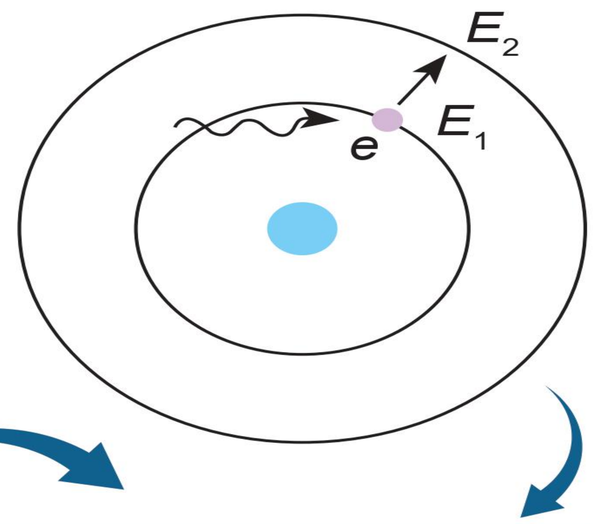
r = radius of orbit

LIMITATIONS

- This model not explain the stability of nucleus.
- This does not explain the line spectra of atom.

POSTULATES

- Electron in an atom could revolve in certain stable orbits with emission of radiant energy.
- $L = \frac{nh}{2\pi}$ L = angular momentum.
  h = Planck's constant = 6.6 × 10–34 JS
- $h\upsilon = Ei - Ef$
  Ei & Ef are the energies of initial & final states. Ei > Ef



EXCITATION ENERGY

$E_{excitation} = E_2 - E_1$

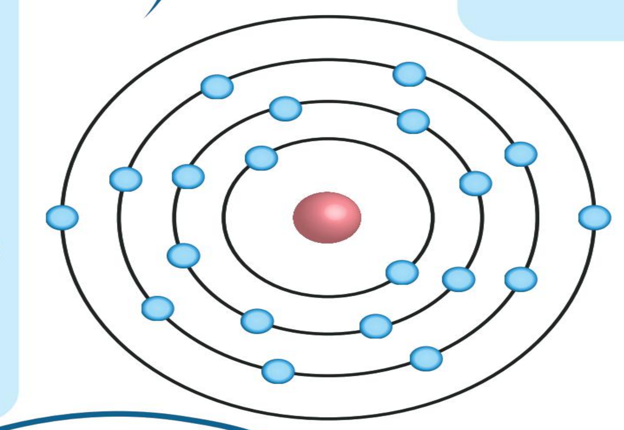
E_1 = energy of lower orbit

E_2 = energy of higher orbit

EXCITATION POTENTIAL

$$V_{excitation} = \frac{E_{excitation}}{e}$$

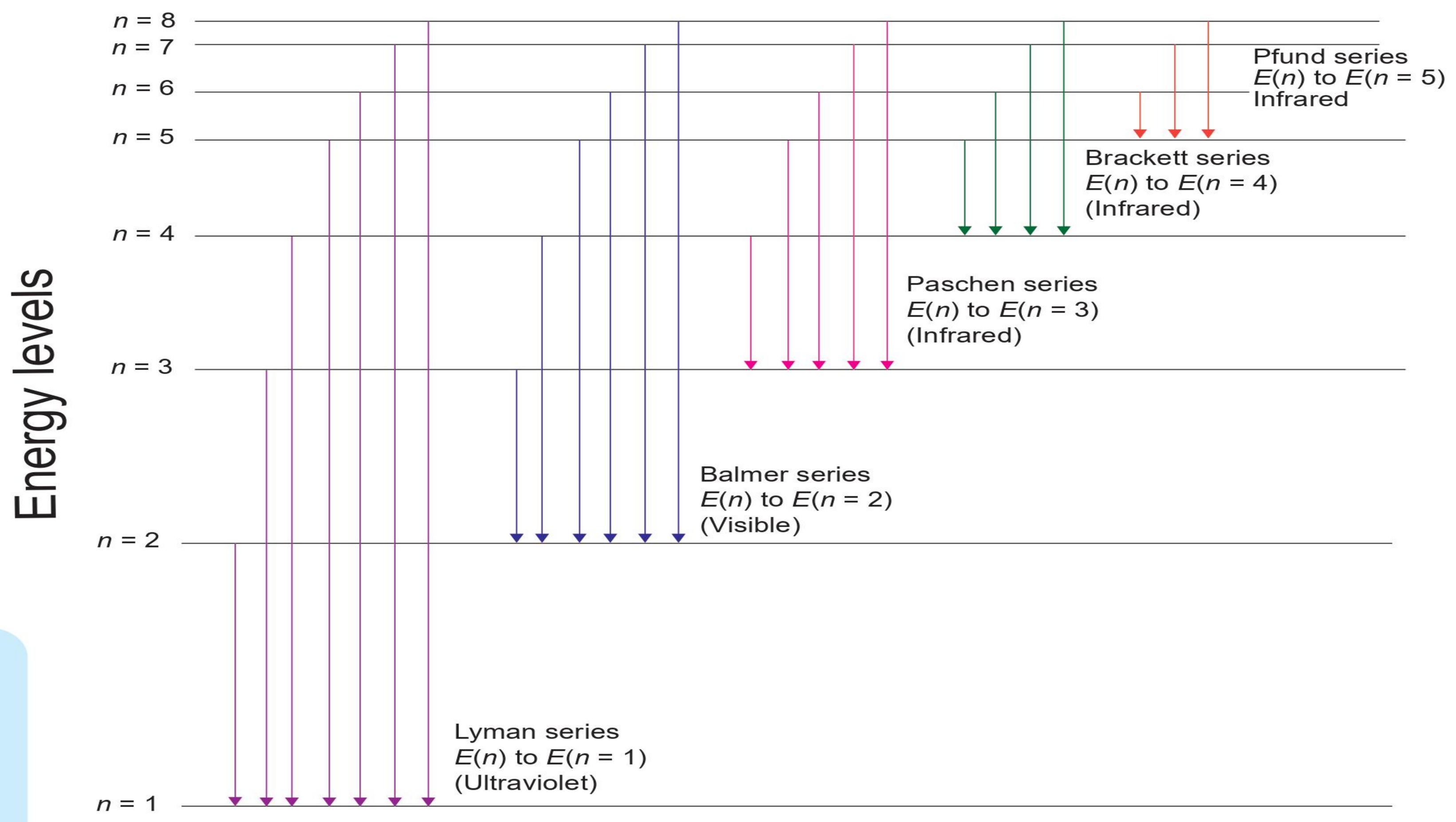
$$= \frac{E_2 - E_1}{e}\text{(volts)}$$



IONIZATION ENERGY

- Minimum energy required to remove the electron.
- $E_{ionization} = \frac{13.6\, z^2}{n^2}$ volts

LINE SPECTRA OF THE HYDROGEN ATOM



(1) The wave number or wavelength of the emitted photon when electron jumps from higher orbital state 'N_2' to lower orbital state 'N_1' is

$$\bar{\nu} = \frac{1}{\lambda} = \frac{En_2 - En_1}{hc} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

R = Rydberg is constant = $1.097 \times 10^7\ m^{-1}$

(2) Number of spectral lines when electron jump from

$$\text{Nth orbit} = \frac{n(n-1)}{2}$$

OUTCOMES

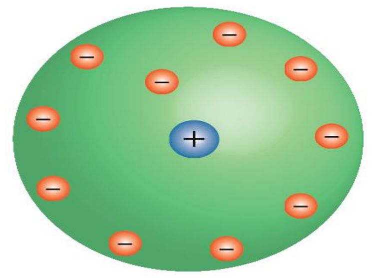
- Most of the α - particles went straight without any deviation.
- Some of α - particles were deflected by some angles.
- Very few α - particles were deflected by an angle 180°

CONCLUSIONS

- Positive charge was concentrated in small region of an atom is called nucleus
- Negative charge were revolving in circular orbit around the nucleus.

RUTHERFORD'S NUCLEAR MODEL OF AN ATOM

- α - particles were emitted by the radioactive element Bi^{214}_{83} & were bombarded on a thin gold foil.
- Scattered α - particles are collected on ZnS screen.



RADIUS OF NTH ORBIT

$$r_n = \frac{n^2h^2\varepsilon_o}{z\pi me^2} = \frac{0.53\, n^2}{z}\ \mathring{A}$$

$$r_n \propto \frac{n^2}{z},\ r_n \propto \frac{1}{m}$$

VELOCITY OF ELECTRON IN NTH ORBIT

$$v_n = \frac{ze^2}{2nh\varepsilon_o} = 2.19 \times 10^6\left(\frac{z}{n}\right)\text{m/s}$$

ORBITAL FREQUENCY IN NTH ORBIT

$$f_n = \frac{v}{2\pi r} = \frac{e^4z^2}{4\varepsilon_o^2n^3h^3}$$

$$f_n \propto \frac{Z^2}{n^3}$$

TOTAL ENERGY IN NTH ORBIT

- $E_N = K_N + U_N = \frac{-me^4z^2}{8\varepsilon_0^2h^2n^2}$
- $E_N = \frac{-13.6z^2}{n^2}$ eV
- $E_N \propto \frac{z^2}{n^2}, E_n \propto m$

POTENTIAL AND KINETIC ENERGY IN NTH ORBIT

$$U_N = \frac{-1}{4\pi\varepsilon_o}\frac{Ze^2}{m} = \frac{me^4z^2}{4\varepsilon_0^2h^2n^2}$$

$$K_N = \frac{1}{2}mv^2 = \frac{me^4z^2}{8\varepsilon_0^2h^2n^2}$$

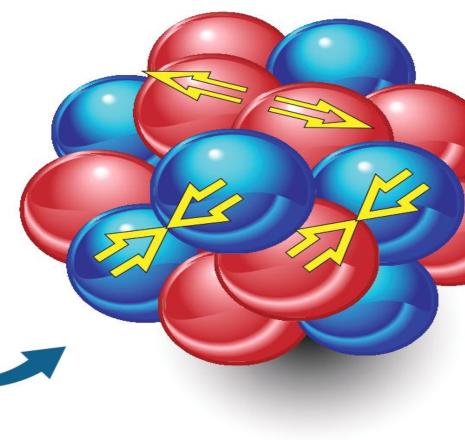
IONIZATION POTENTIAL

- $V_{ionization} = \frac{E_{ionization}}{e}$

$$= \frac{13.6\ z^2}{n^2}\text{volts}$$

BINDING ENERGY

- Minimum energy required to bound the electron from nucleus.
- $B.E. = -E_{ionization} = \frac{-13.6\ z^2}{n^2}$ eV



ATOMS

CHAPTER TWELVE
CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

August 28, 2025

Learning Objectives:

- Introduction
- Alpha-particle scattering experiment;
- Rutherford’s model of atom;
- Bohr model of hydrogen atom, Expression for radius of nth possible orbit, velocity and energy of electron in nth orbit,
- Hydrogen line spectra (qualitative treatment only).

INTRODUCTION TO ATOMS

Atomic Hypothesis

- By the 19th century, enough evidence supported the **atomic hypothesis of matter**.

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- In 1897, **J. J. Thomson** discovered negatively charged constituents, called **electrons**, identical for all atoms.
- Atoms are overall **electrically neutral** $\Rightarrow$ must contain **positive charge** too.
- The key question: How are **positive charges and electrons** arranged inside the atom?

THOMSON’S MODEL (1898)

Plum Pudding Model

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- Proposed by **J. J. Thomson** in 1898.
- Atom as a **uniformly distributed positive sphere**.
- Electrons embedded like **seeds in a watermelon**.
- Known as the **plum pudding model**.
- **Limitation:** Later experiments showed actual distribution of charges was very different.

SPECTRAL STUDIES

Discrete Spectra

- **Condensed matter** emits a **continuous spectrum**.

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- **Rarefied gases** (heated or excited) emit radiation of **discrete wavelengths only**.

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- Each element has a **unique spectrum**.
- Example: **Hydrogen spectrum** $\Rightarrow$ fixed set of lines, always same relative positions.
- **J. J. Balmer (1885)** gave empirical formula for wavelengths of hydrogen spectrum.

RUTHERFORD'S WORK

Nuclear Model of Atom

- **Ernest Rutherford (1906–1911)** studied α-particle scattering.

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- Result $\Rightarrow$ Positive charge and most of mass concentrated in a **small nucleus**.
- Electrons revolve around nucleus like **planets around Sun**.
- This is the **planetary or nuclear model**.

LIMITATIONS OF RUTHERFORD’S MODEL

Unanswered Questions

- Why are atoms **stable**? (Classical theory predicts electrons should spiral into nucleus.)

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- Why are atoms **stable**? (Classical theory predicts electrons should spiral into nucleus.)
- Why do atoms emit **only discrete line spectra**, not a continuous one?
- These limitations motivated the need for a new model $\Rightarrow$ **Bohr's Model**.

ALPHA-PARTICLE SCATTERING AND RUTHERFORD'S NUCLEAR MODEL OF ATOM

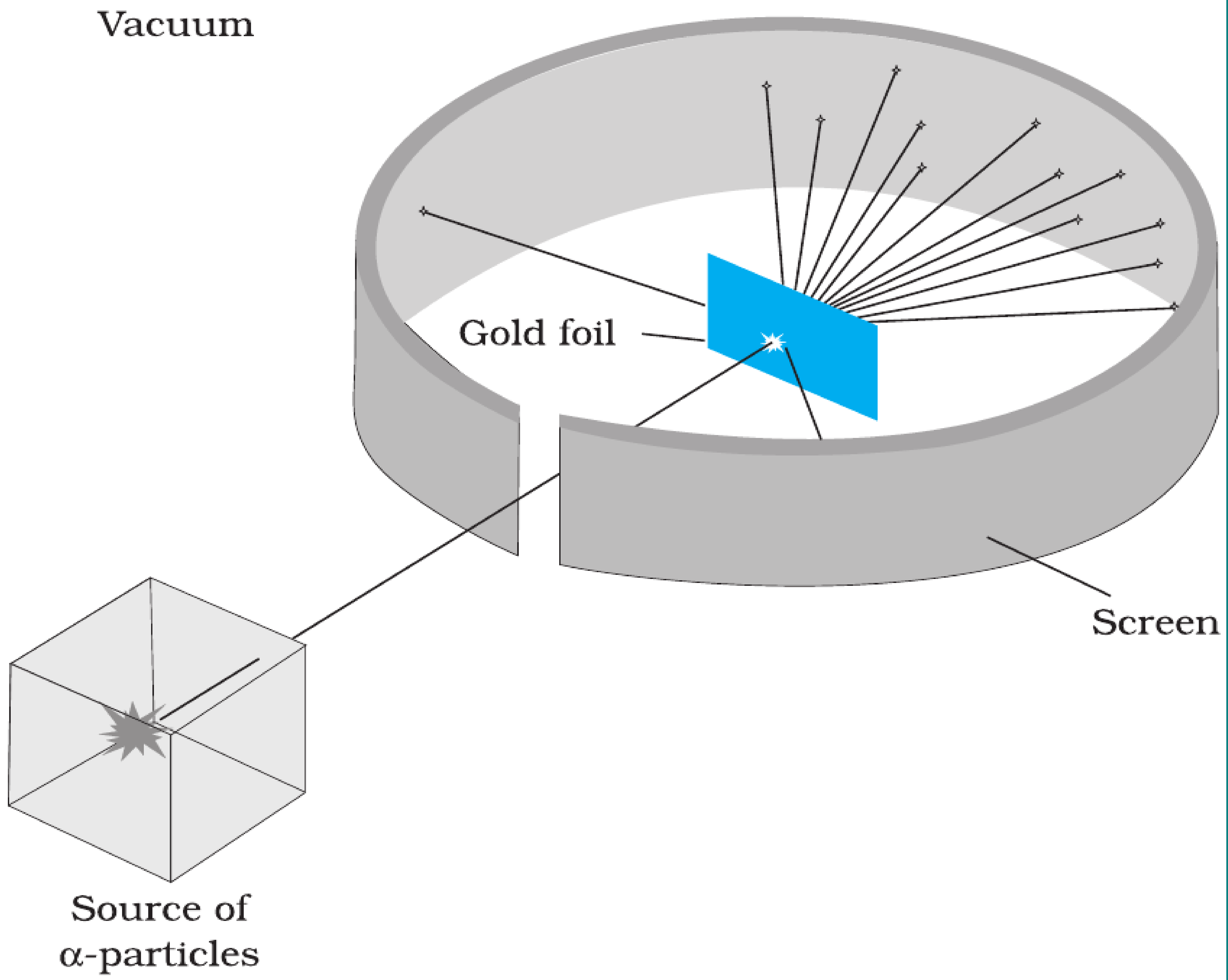


FIGURE 12.1 Geiger-Marsden scattering experiment. The entire apparatus is placed in a vacuum chamber (not shown in this figure).

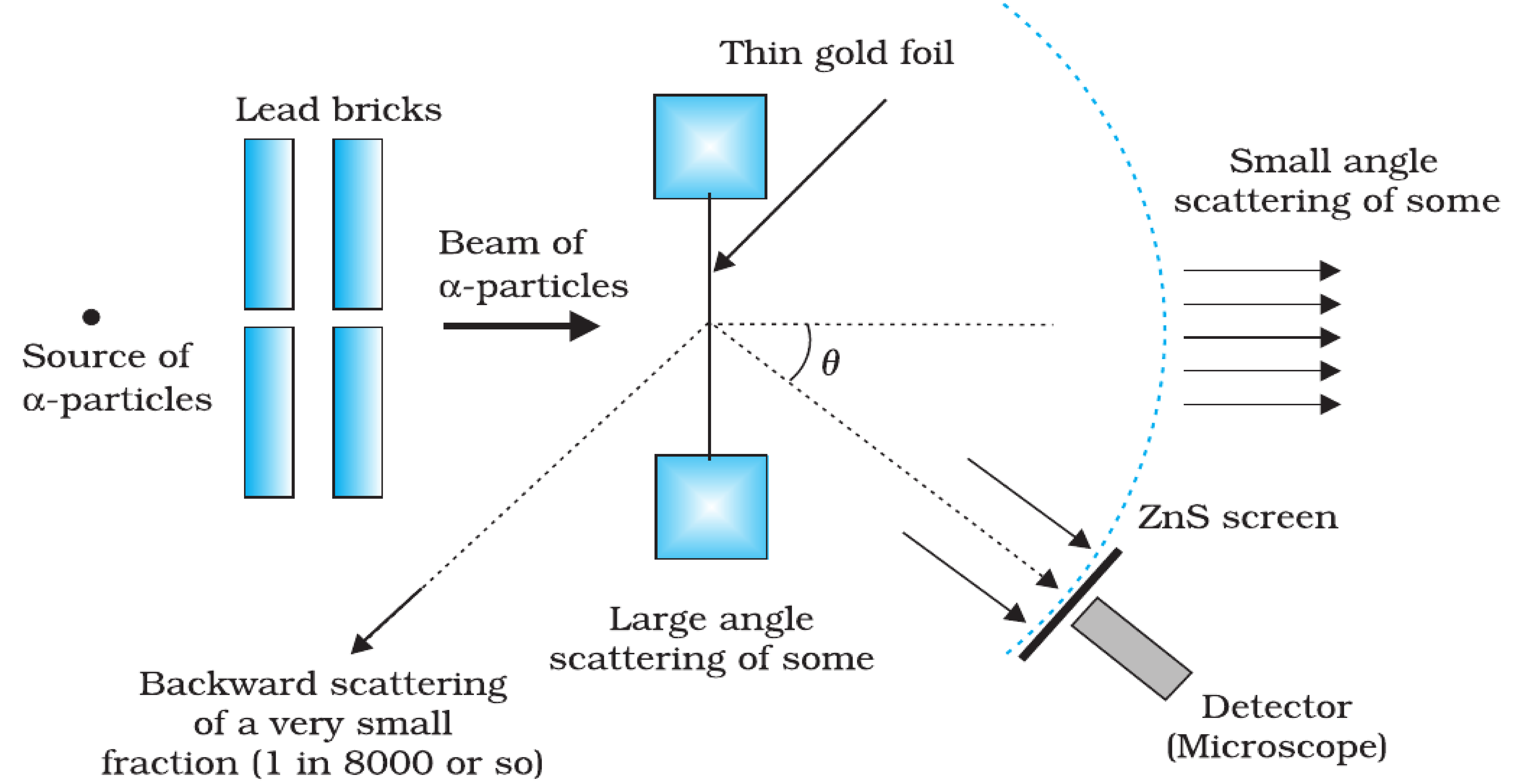
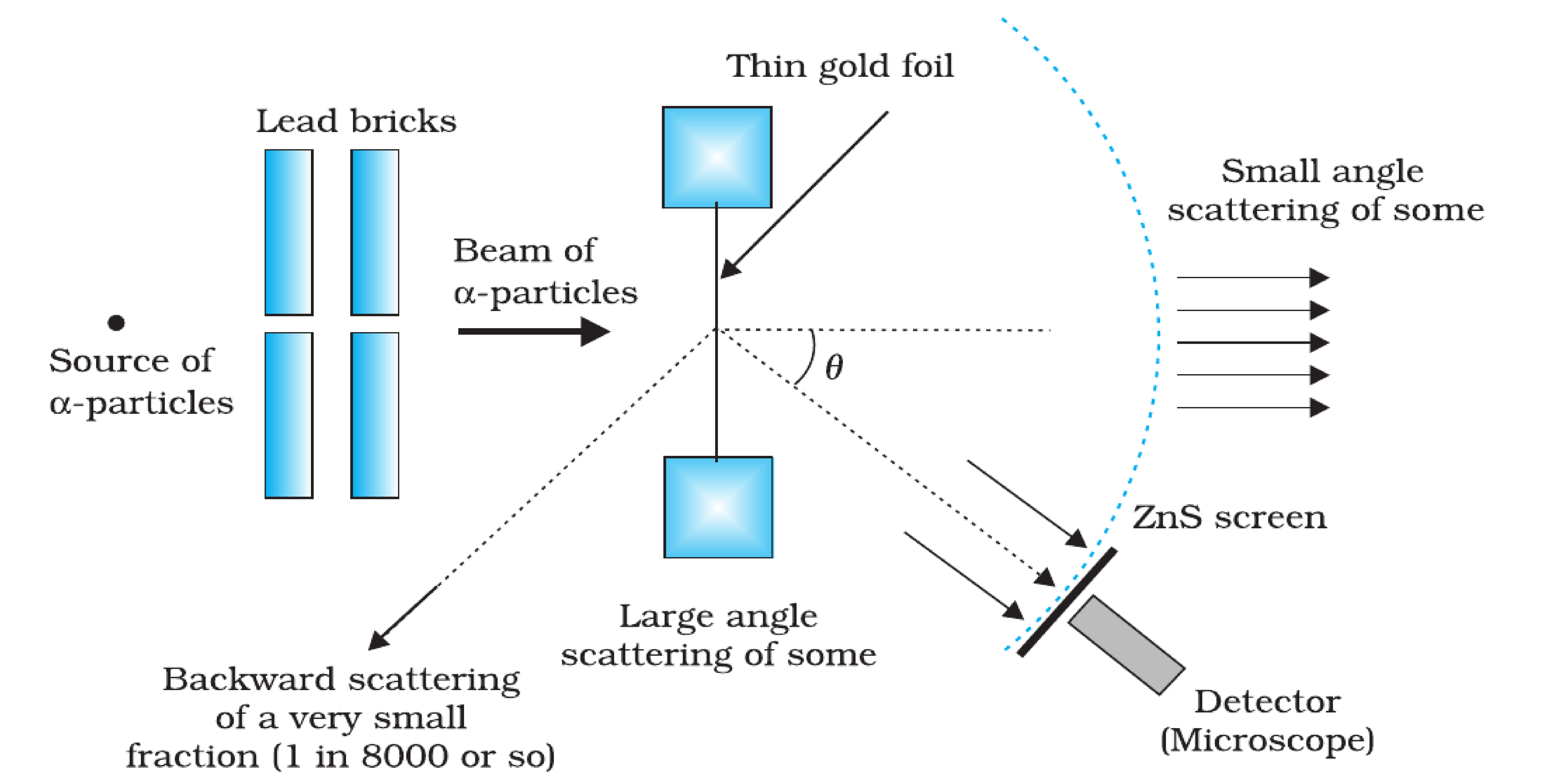


FIGURE 12.2 Schematic arrangement of the Geiger-Marsden experiment.

GEIGER–MARSDEN EXPERIMENT (1911)



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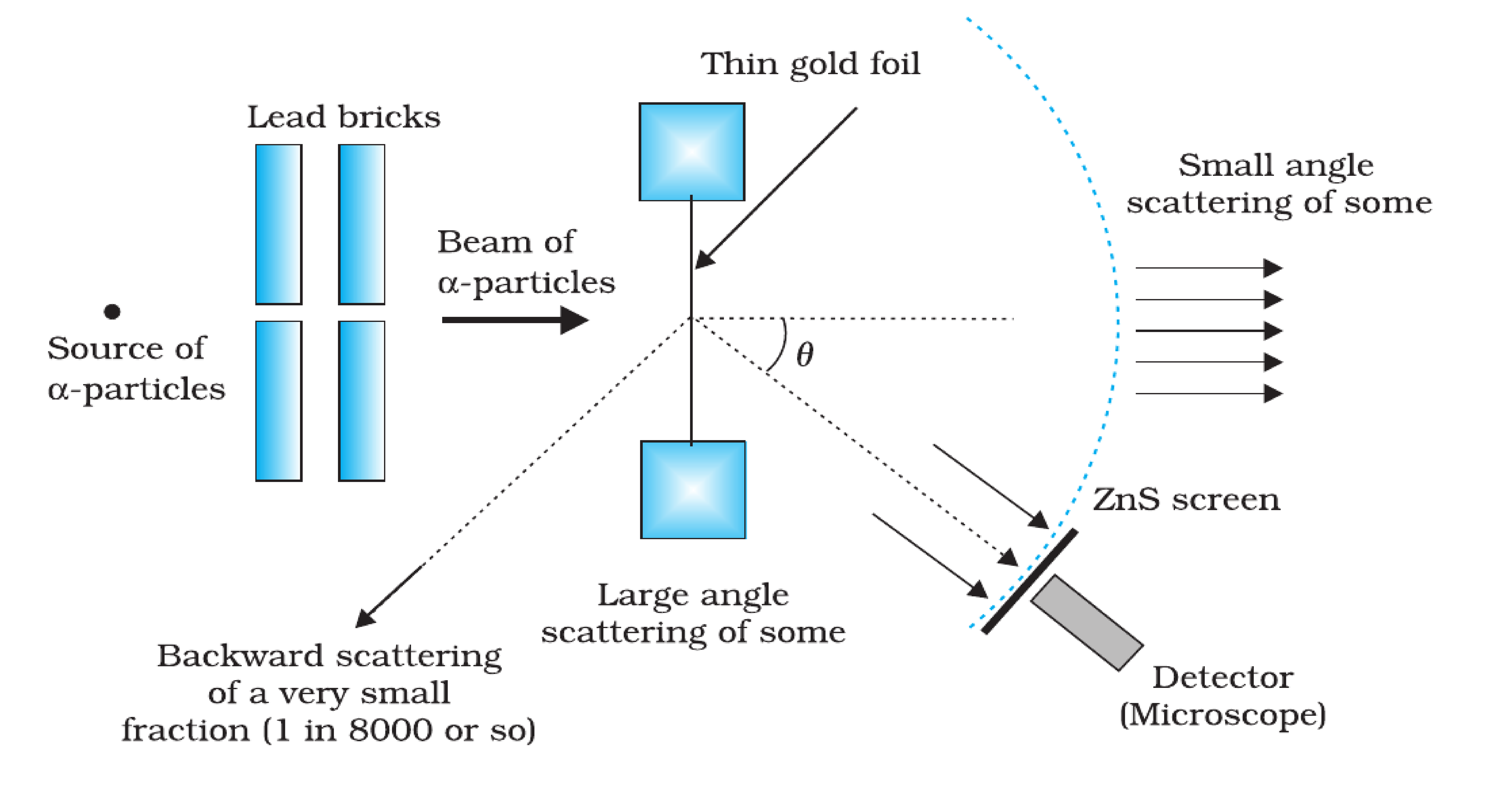

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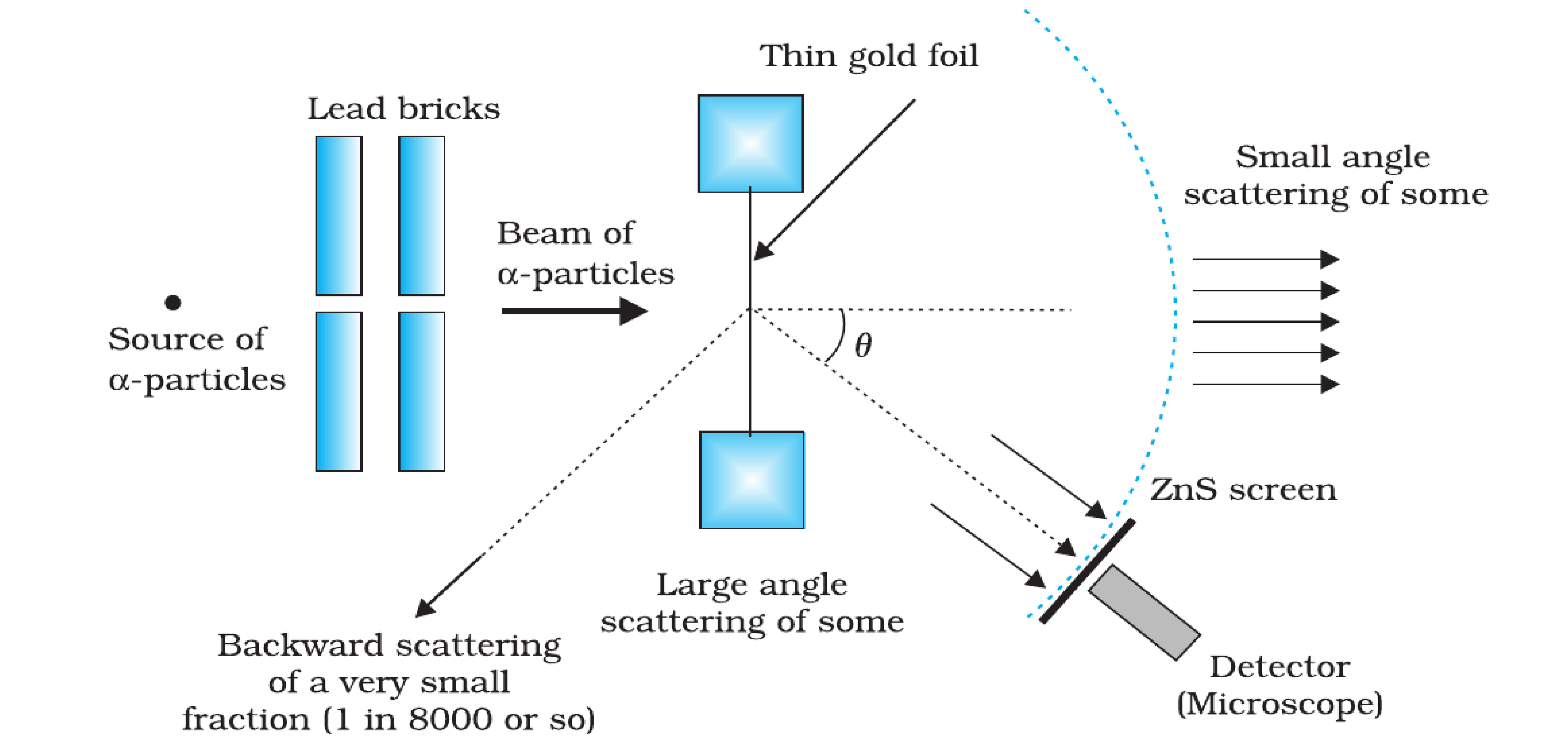

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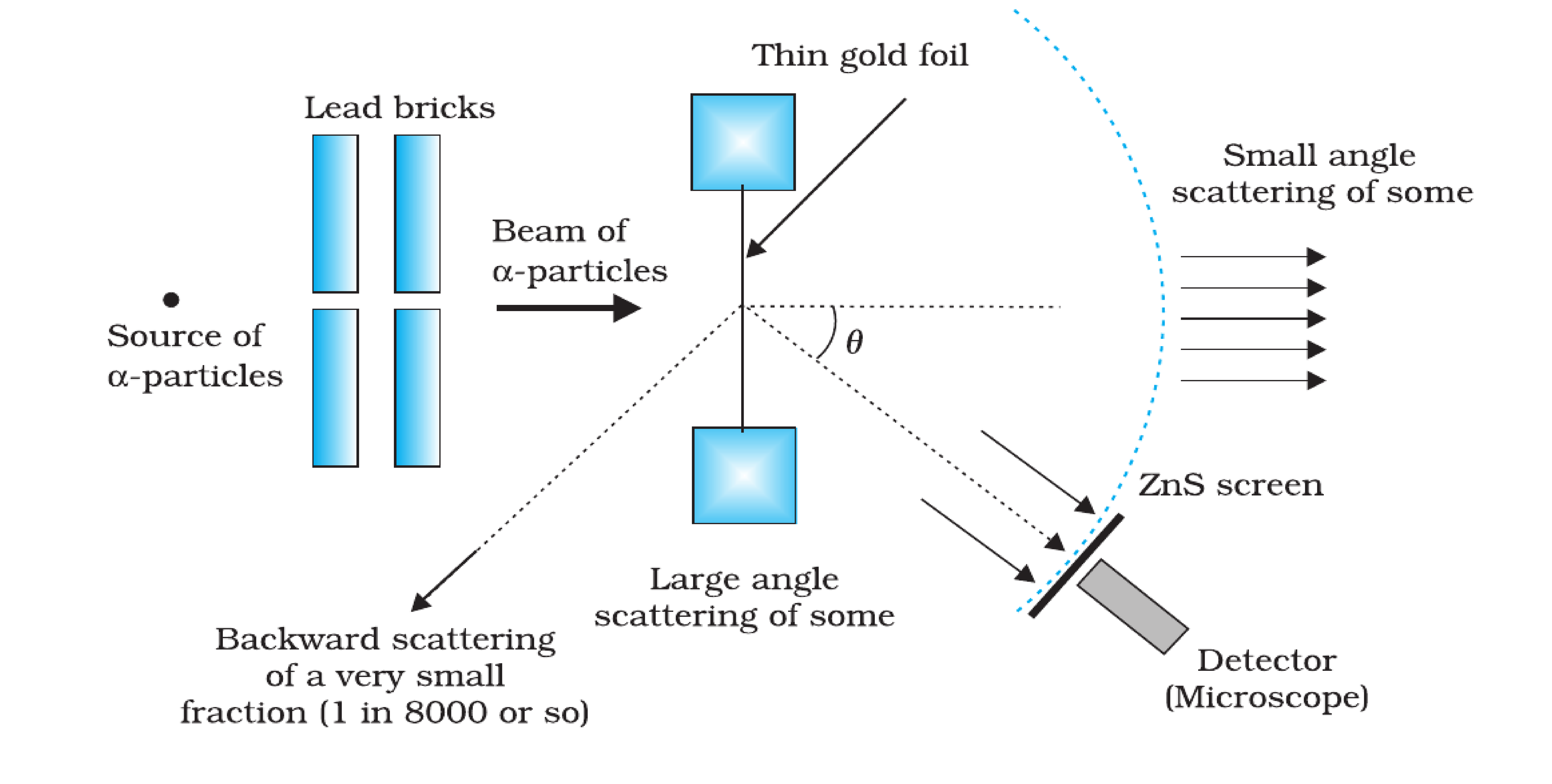

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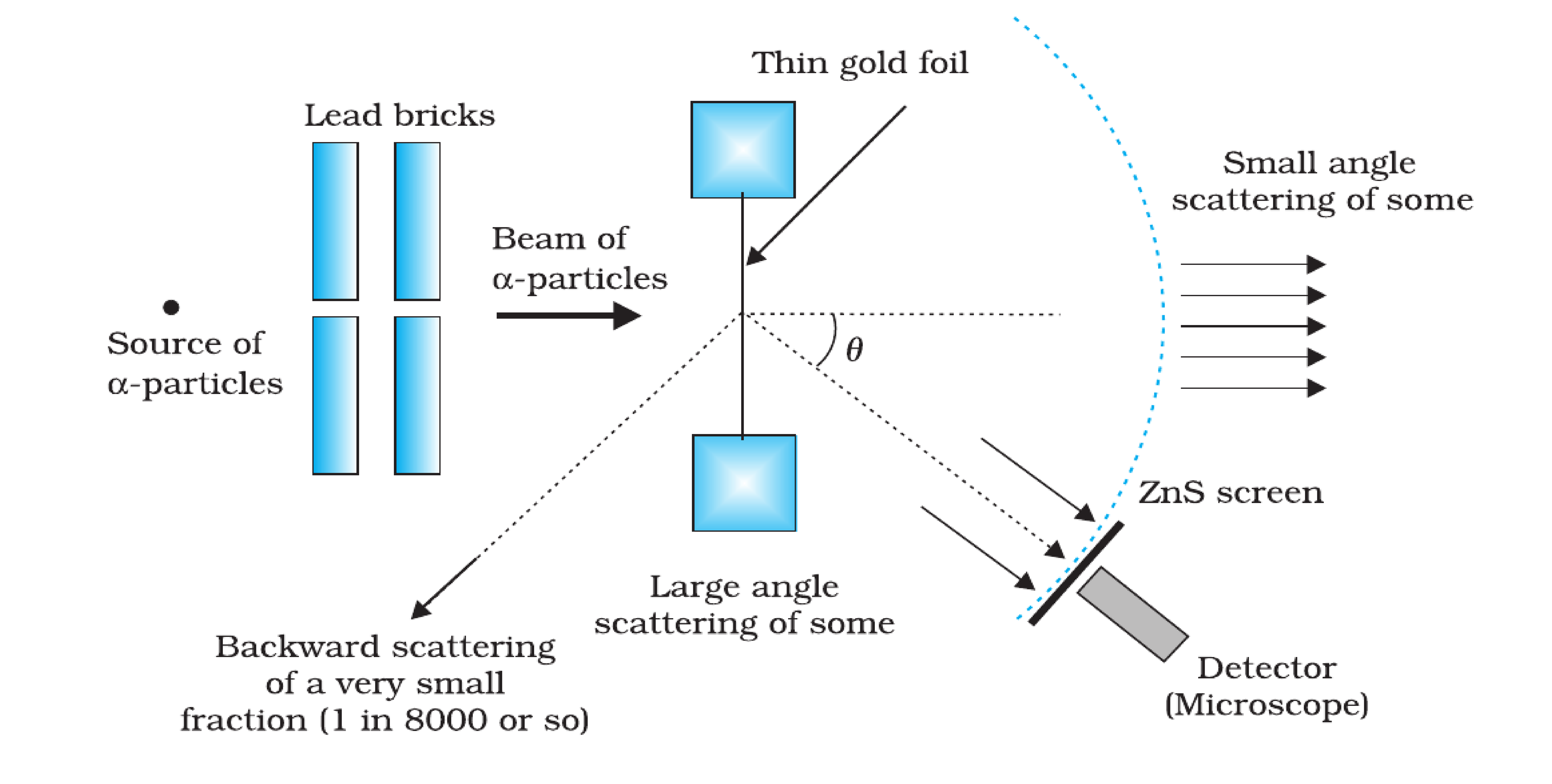

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- Entire setup kept in a **vacuum chamber**.

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Key Experimental Findings

- $\sim$99% α-particles passed through without any deflection.

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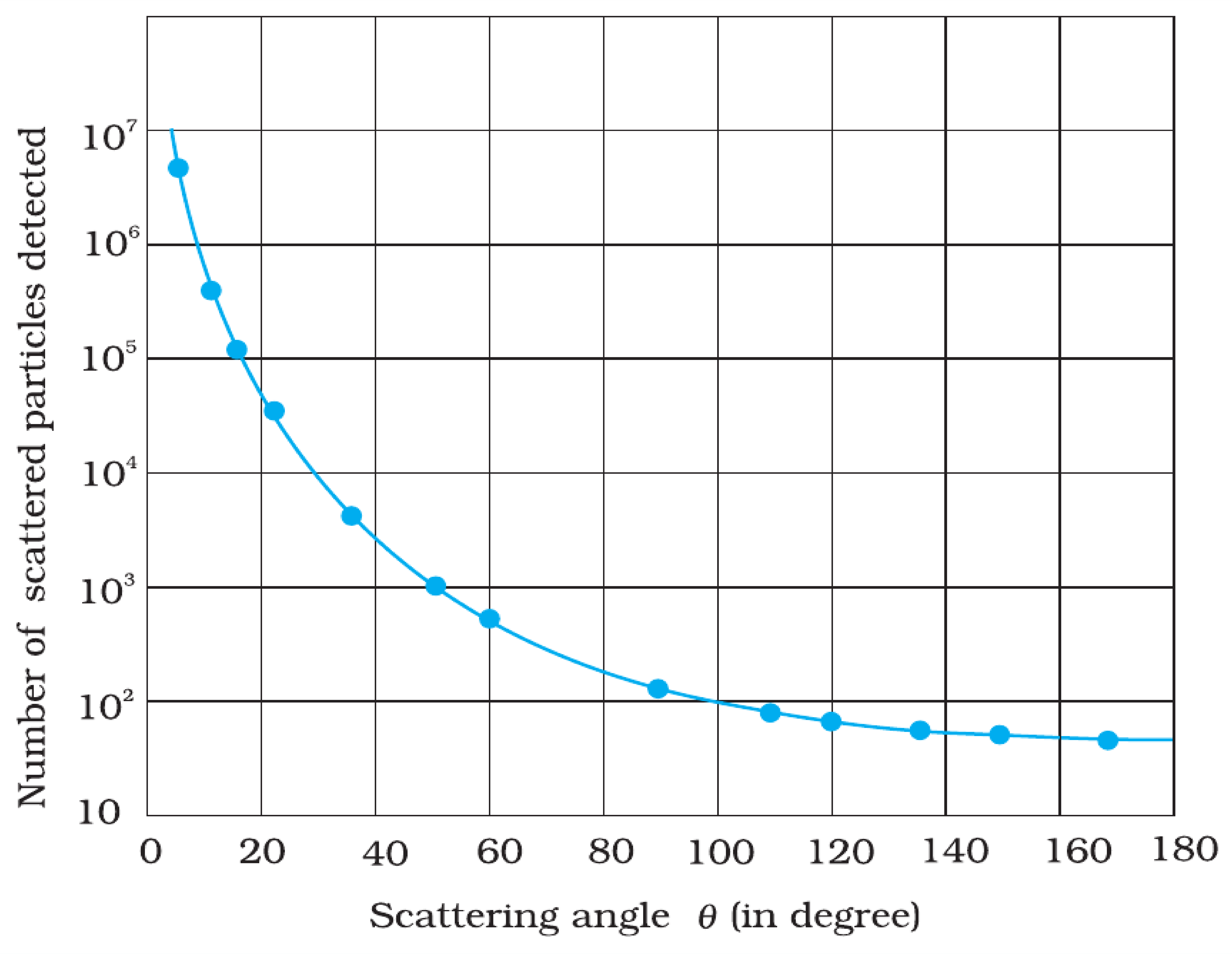
- About **0.14%** deflected by small angles ($> 1^\circ$).
- About **1 in 8000** deflected at large angles ($> 90^\circ$), some even bounced back.
- Such large deflections were **impossible to explain** with Thomson's model of atom.

GRAPHICAL RESULTS



Scattering Curve

- Plot: **Number of scattered α-particles vs. scattering angle θ.**

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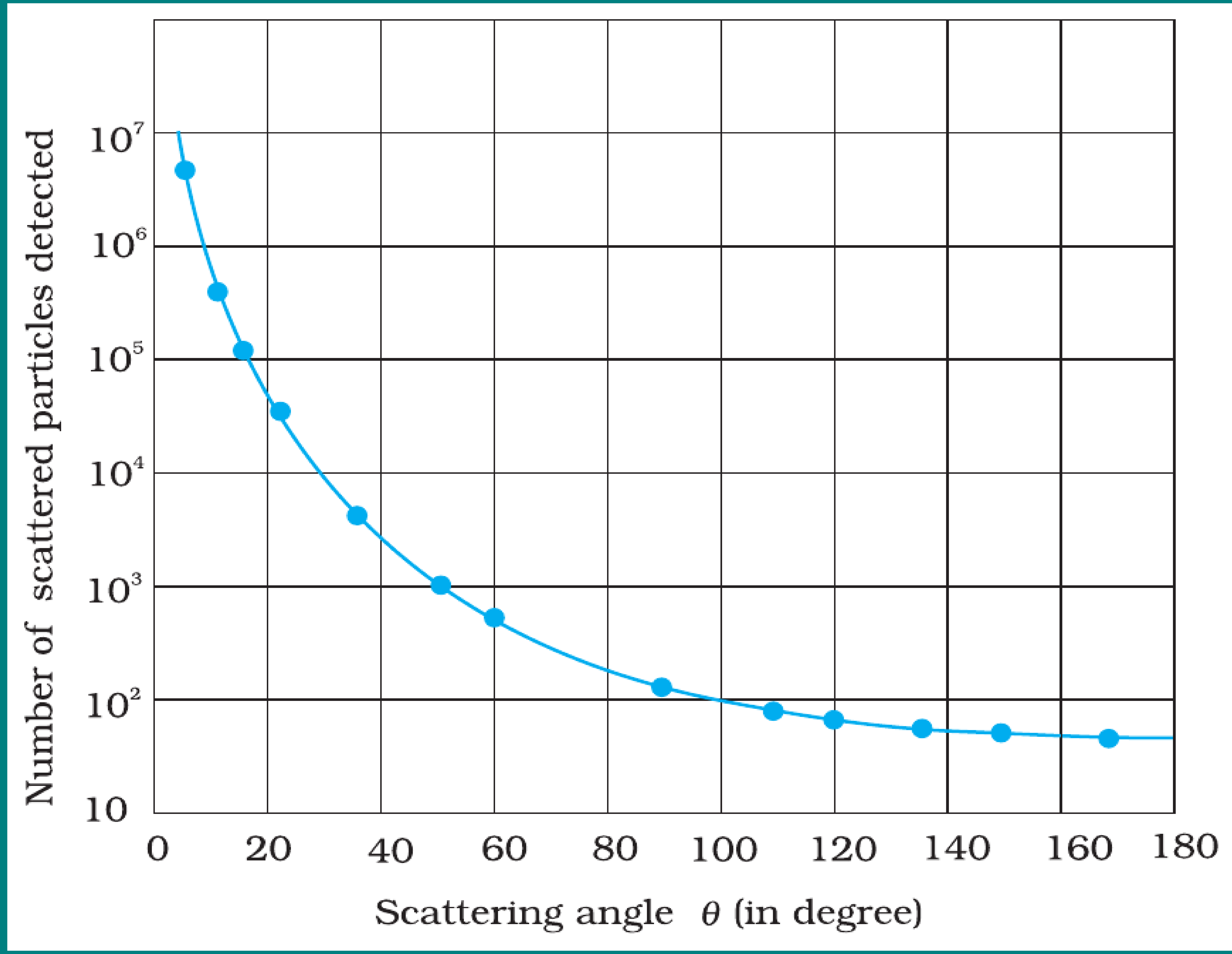

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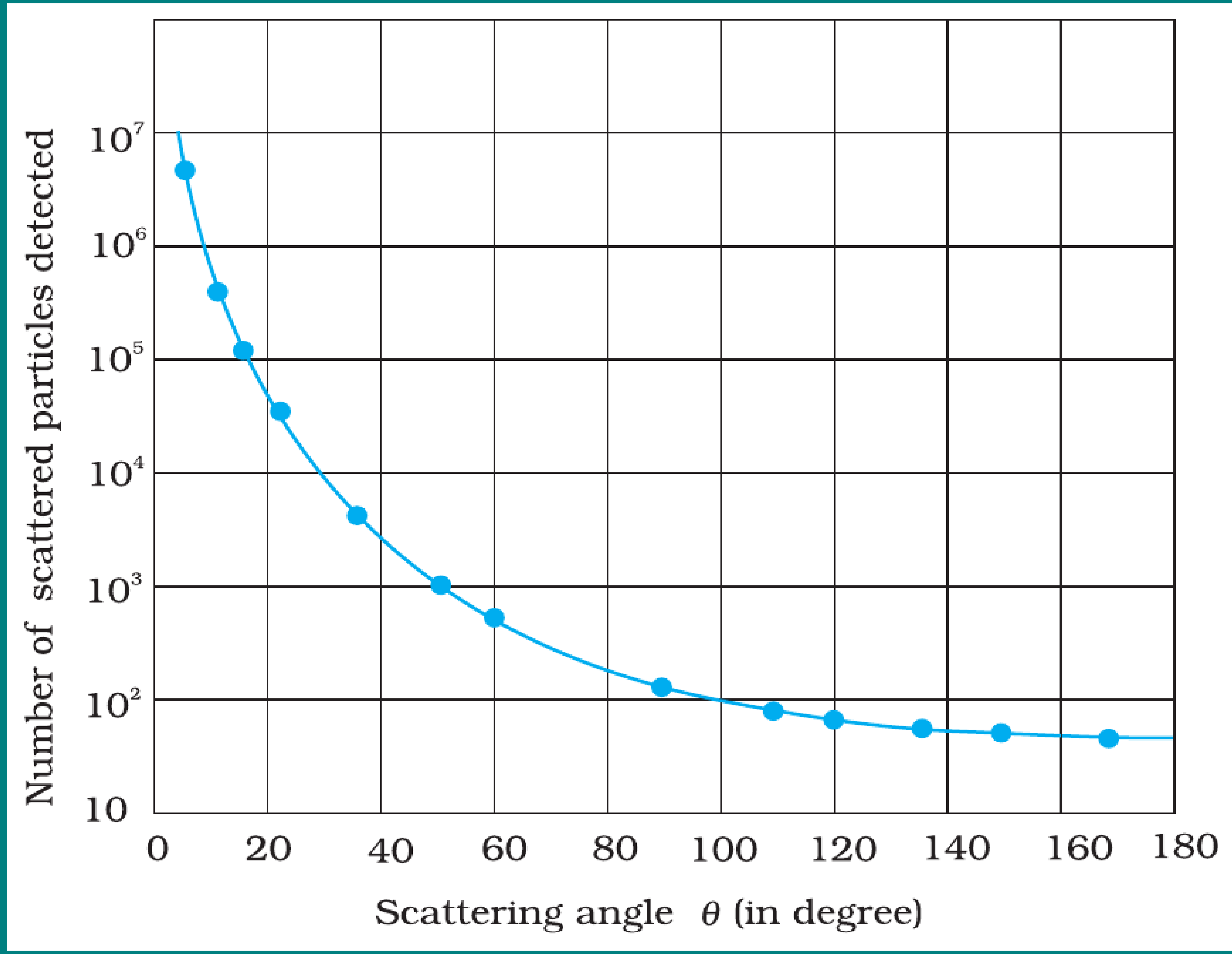

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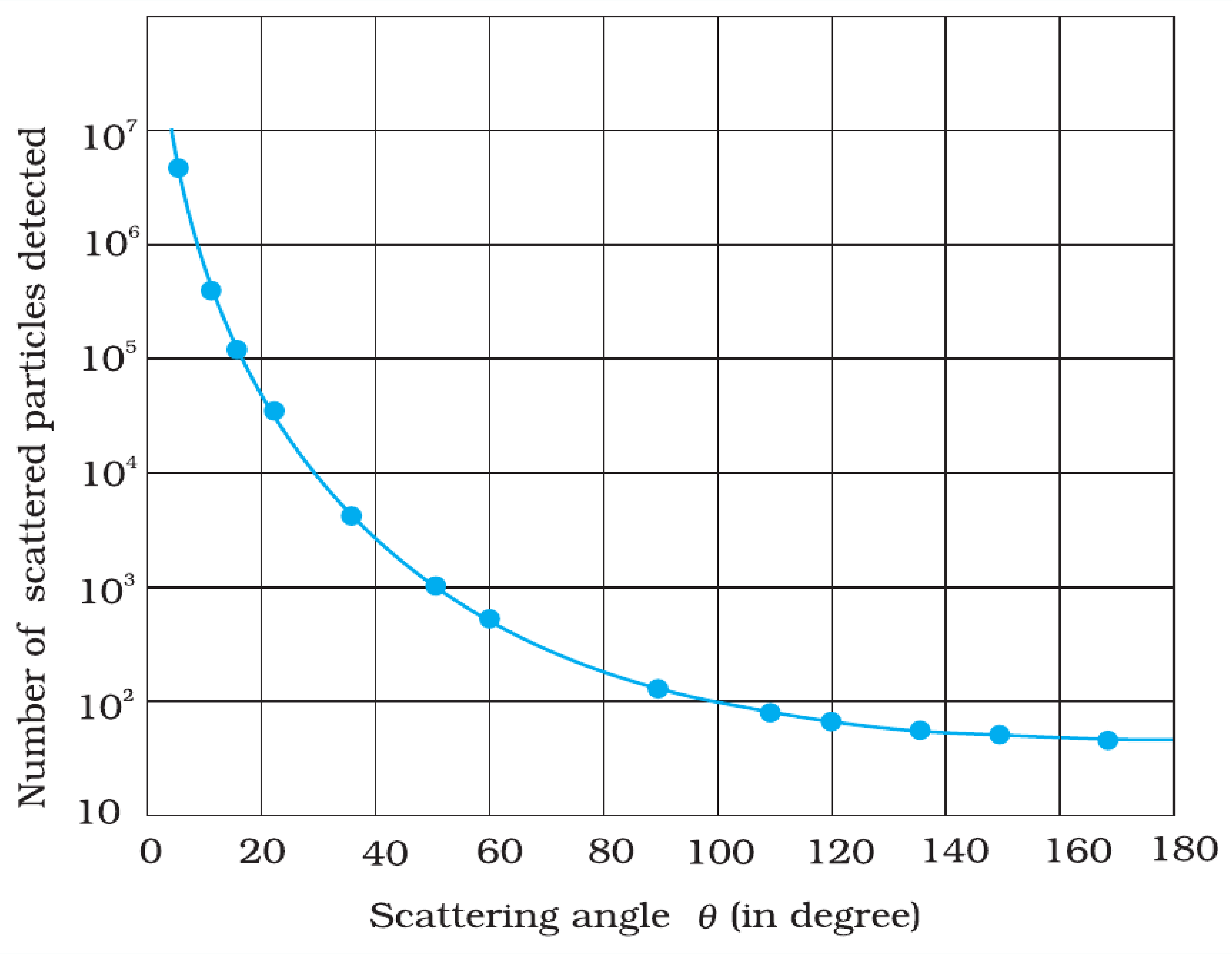

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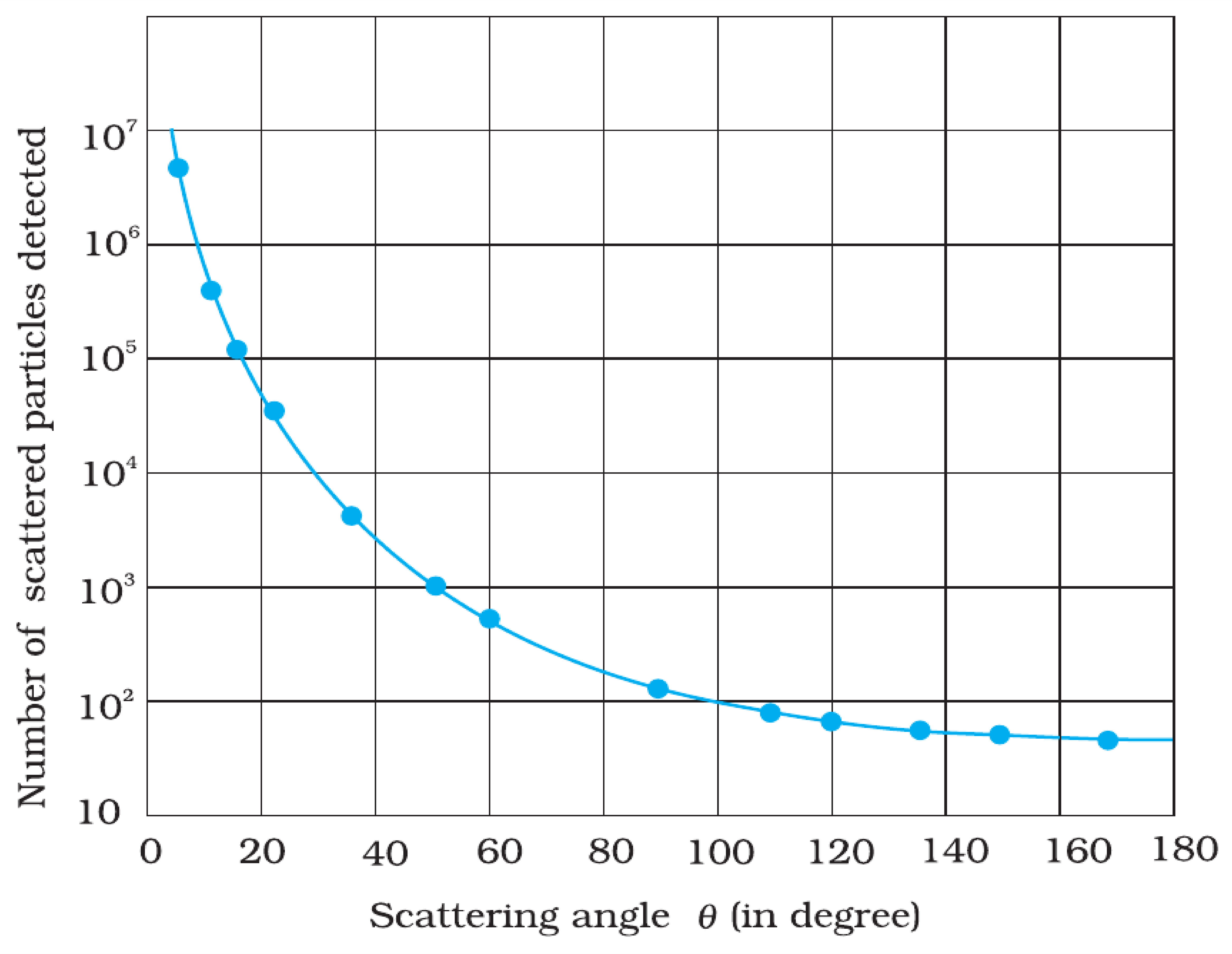

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- This confirmed: **positive charge is concentrated in a small, dense nucleus**.

RUTHERFORD’S NUCLEAR MODEL

Features of the Model

- Atom has a tiny, dense, **positively charged nucleus** at the centre.

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- Nearly all the **mass of atom** is concentrated in the nucleus.
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- Most of the atom is **empty space** $\Rightarrow$ explains why most α-particles passed through undeflected.

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- These limitations paved the way for **Bohr's Model of Atom**.

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Force Expression

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- Becomes very large when r is very small $\Rightarrow$ responsible for large-angle deflections.

TRAJECTORY OF α-PARTICLES

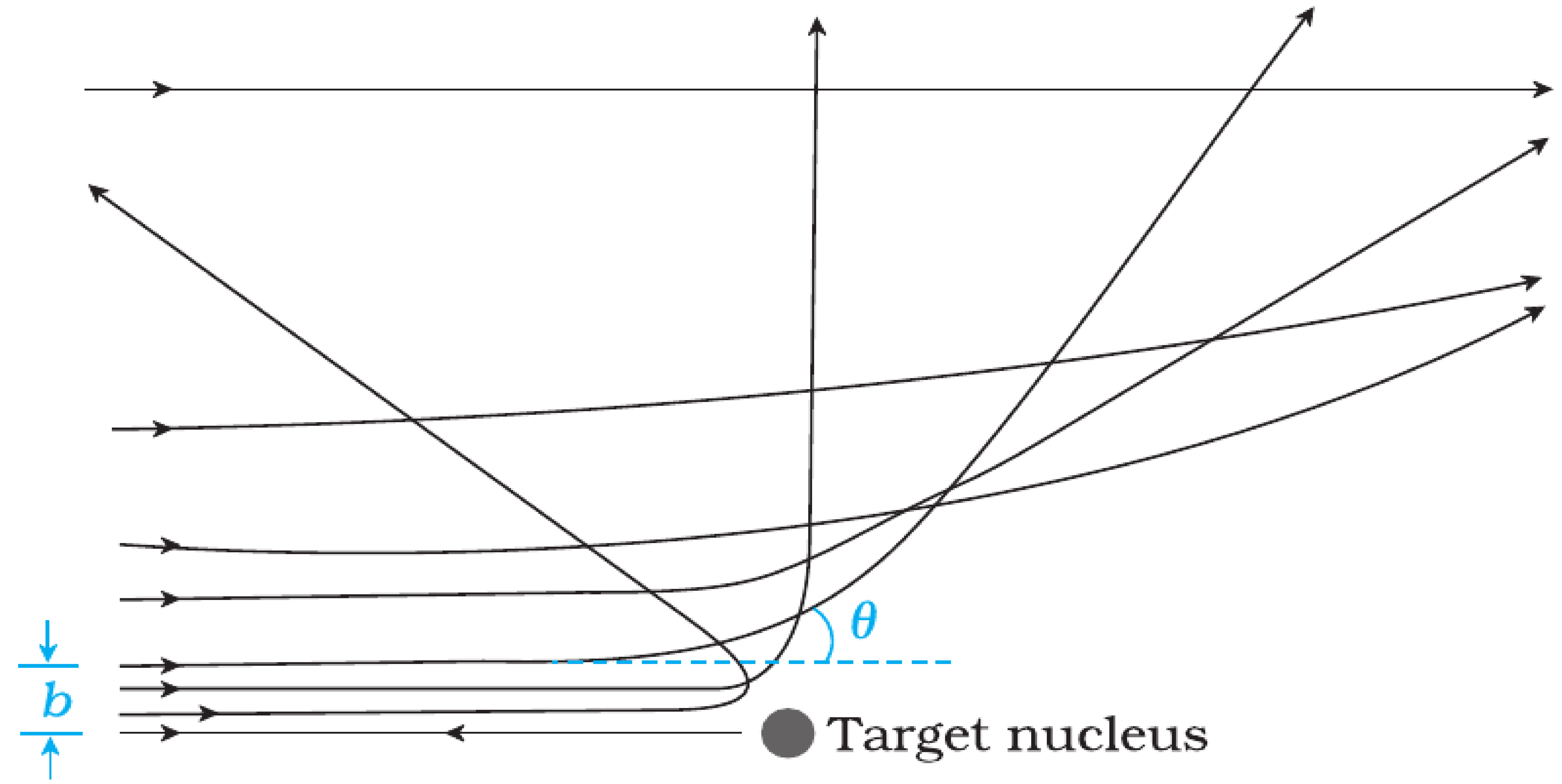

FIGURE 12.4 Trajectory of α-particles in the coulomb field of a target nucleus. The impact parameter, b and scattering angle θ are also depicted.

Deflected Paths

- Far from nucleus: force negligible, path is straight line.

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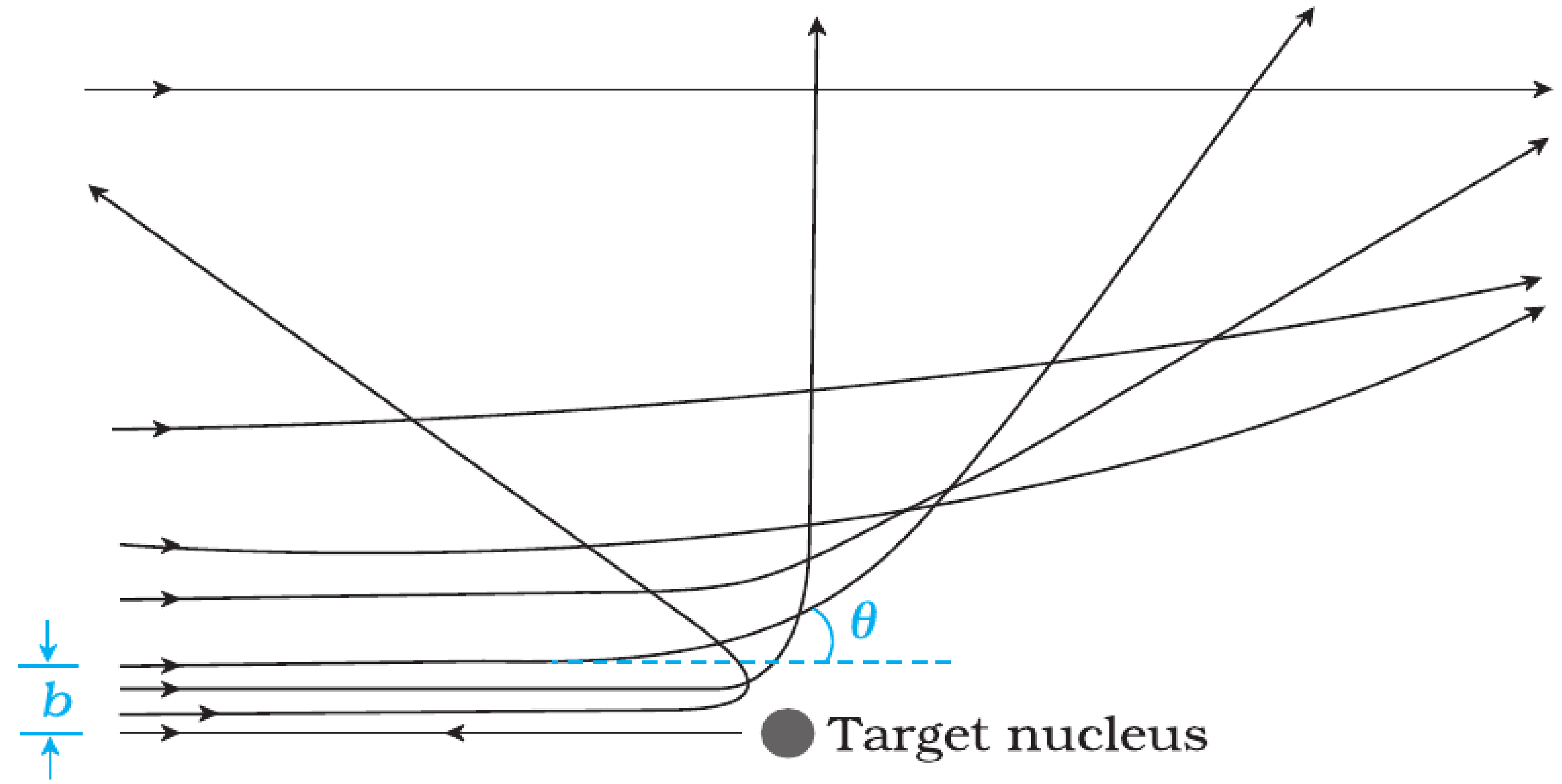

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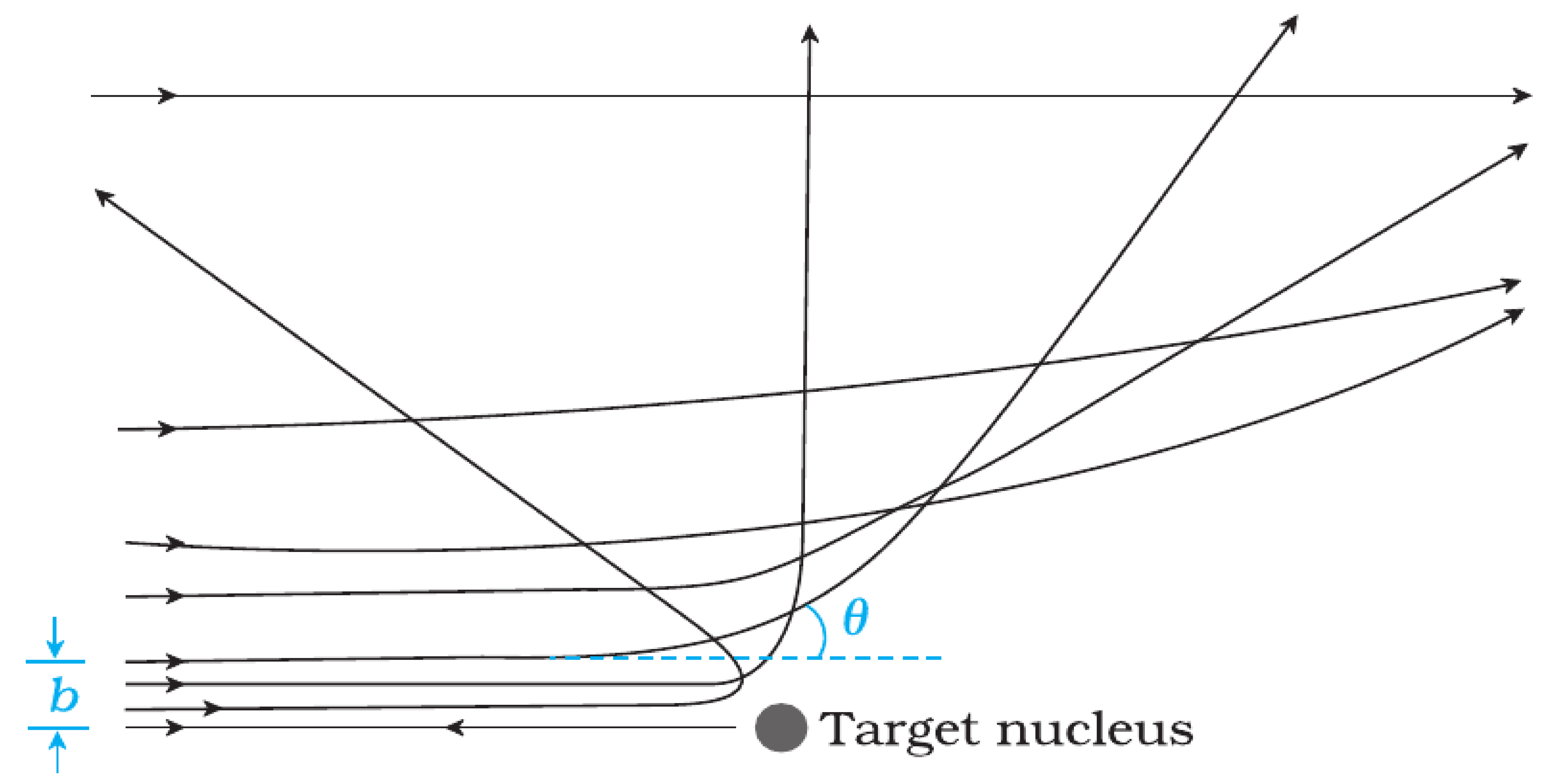

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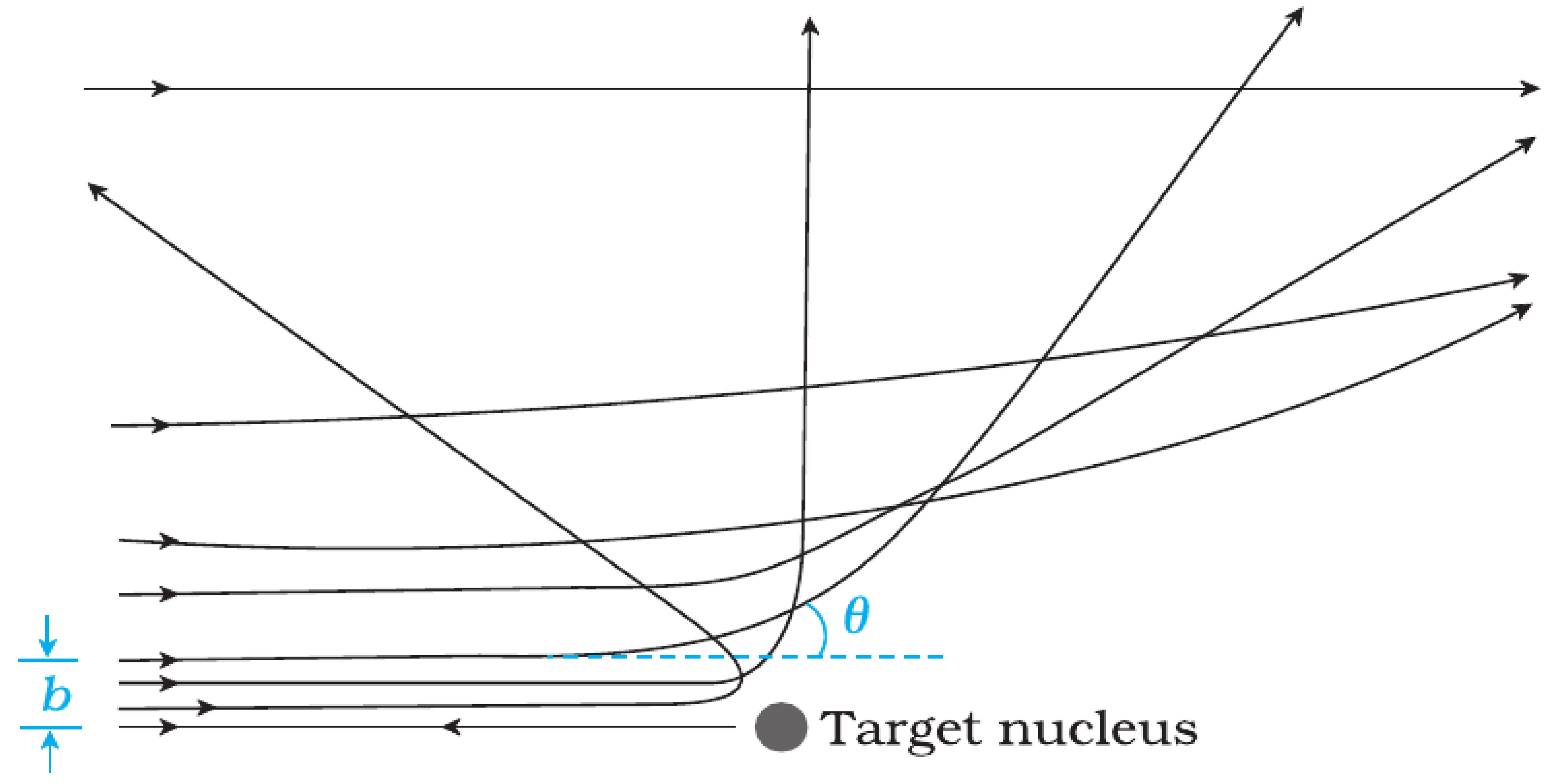

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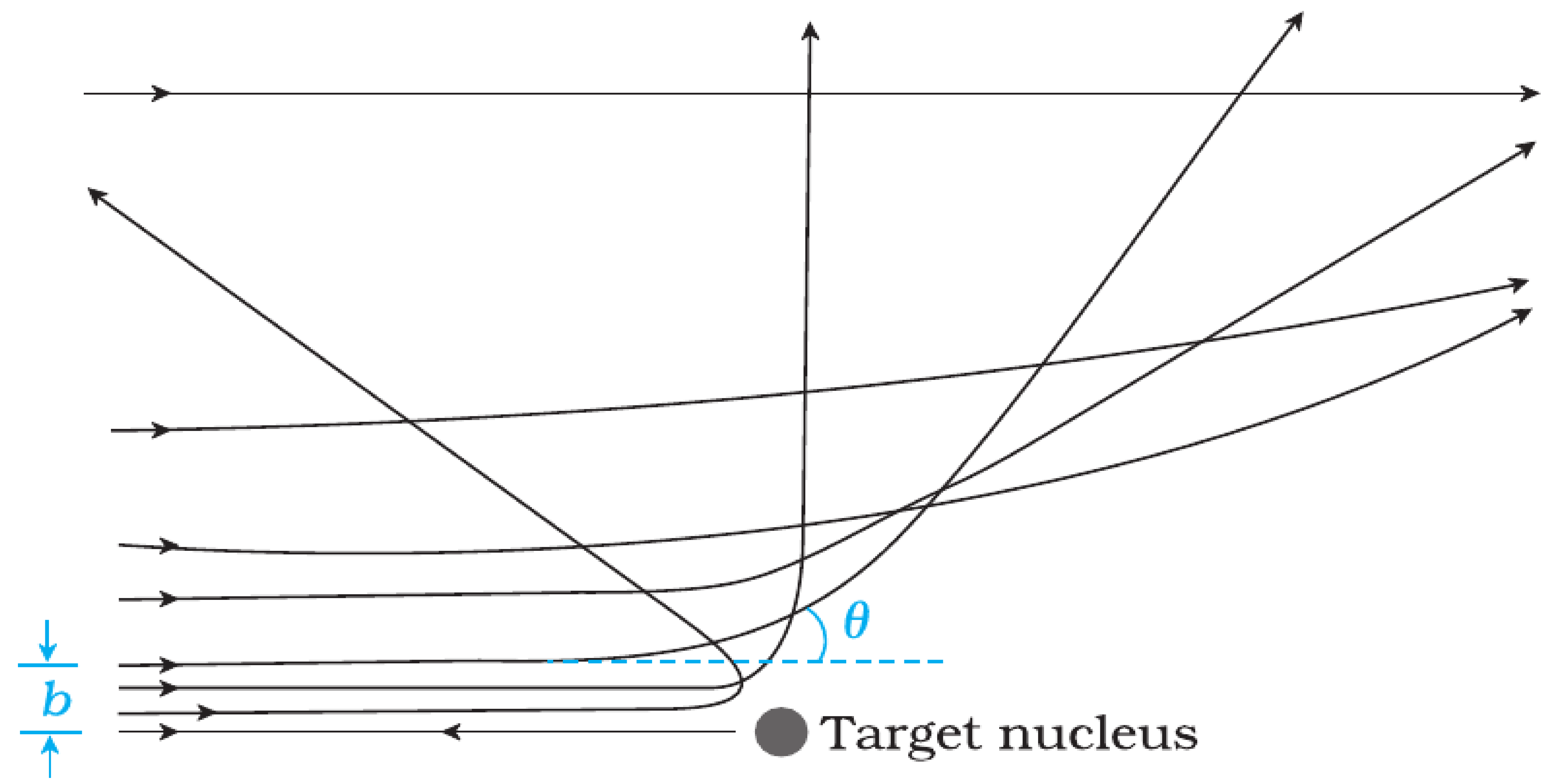
- **Large deflections/backscattering:** very small distance of approach.

IMPACT PARAMETER (b)



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Definition & Role

- **Impact parameter b:** perpendicular distance between initial velocity vector of α-particle (undeflected path) and nucleus centre.

IMPACT PARAMETER (b)

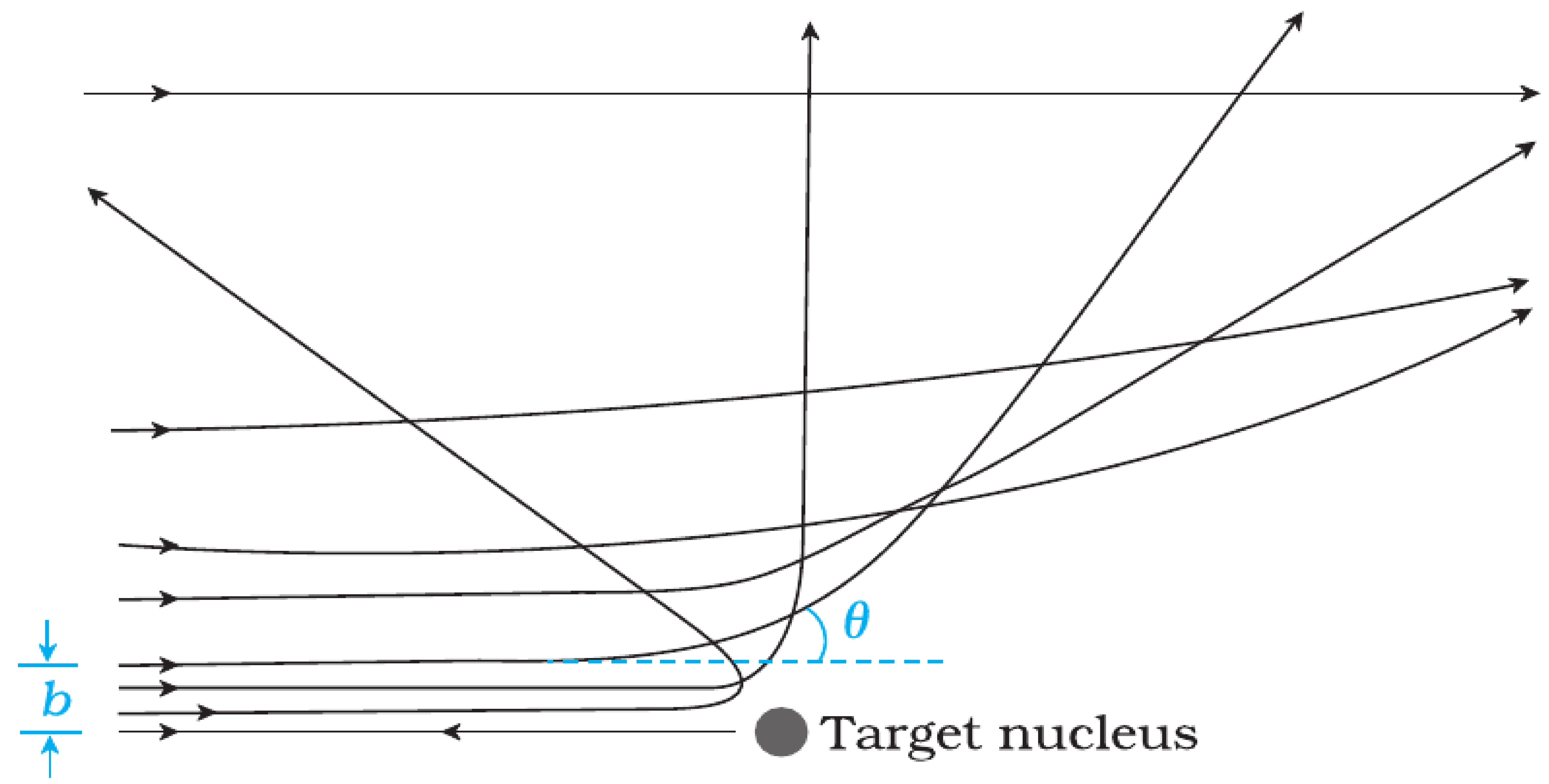

FIGURE 12.4 Trajectory of α-particles in the coulomb field of a target nucleus. The impact parameter, b and scattering angle θ are also depicted.

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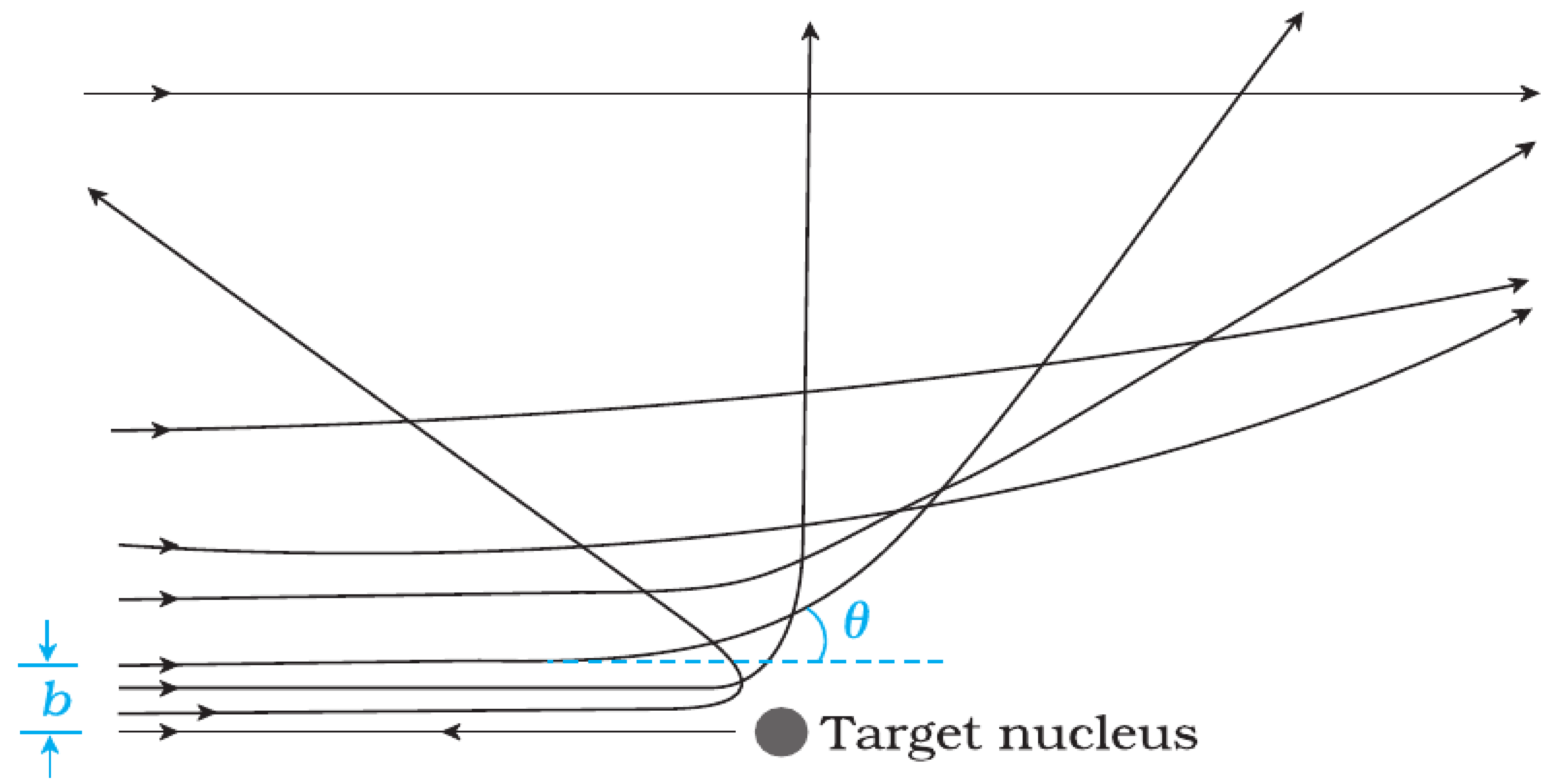

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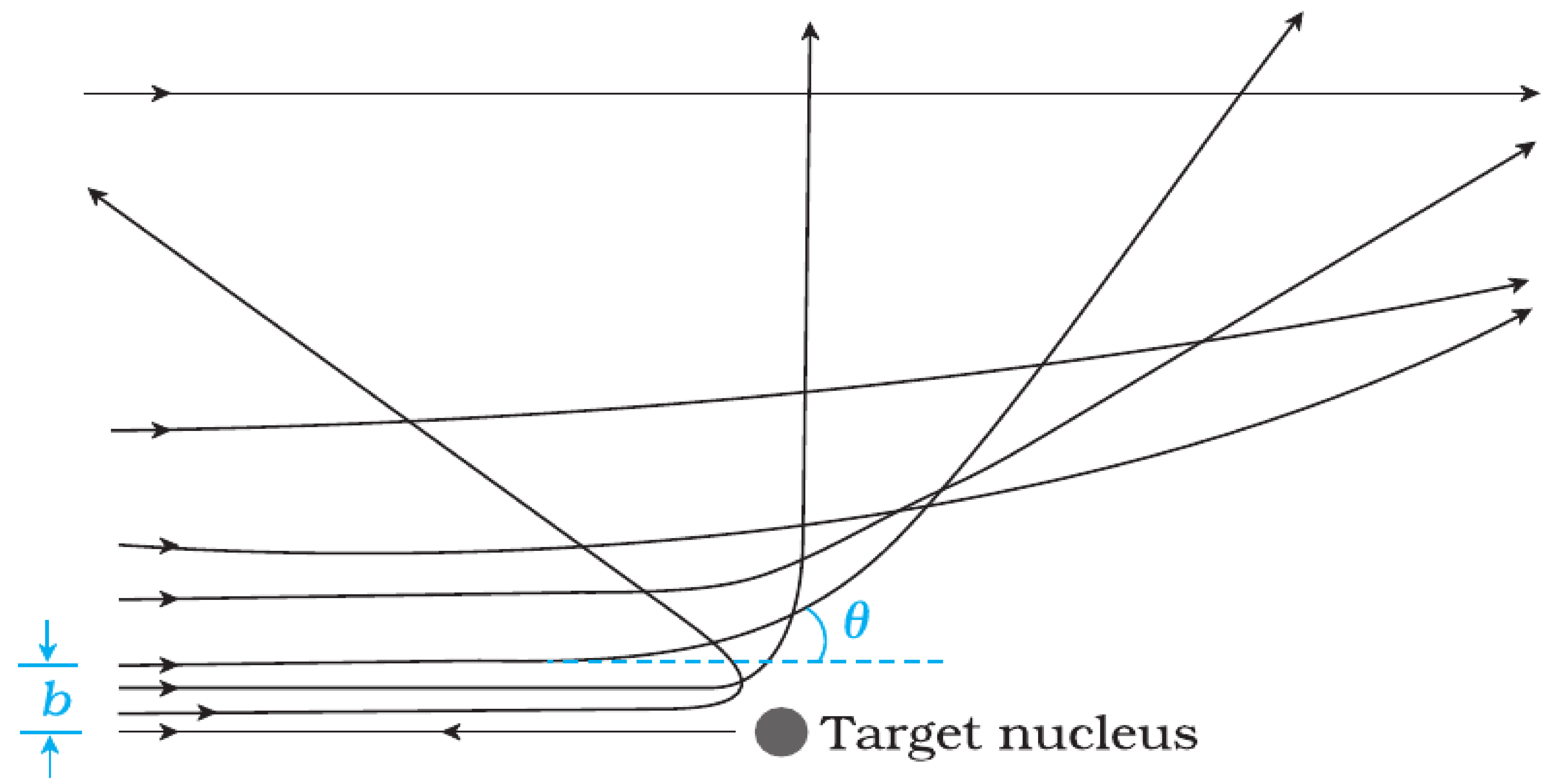

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RELATION BETWEEN b AND SCATTERING ANGLE θ

Mathematical Relation

- Rutherford derived the following relation:

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  - For very small b, $\theta > 90^\circ$ (backscattering possible).

DISTANCE OF CLOSEST APPROACH (r_{min})

Concept

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- For gold nucleus ($Z = 79$) with 5.5 MeV α-particles, $r_{\min} \sim 3.2 \times 10^{-14}$ m.
- Since actual nuclear radius is smaller, the experiment proved that **nucleus is extremely small compared to atom**.

Example 12.1 In the Rutherford's nuclear model of the atom, the nucleus (radius about 10^{-15} m) is analogous to the sun about which the electron move in orbit (radius $\approx 10^{-10}$ m) like the earth orbits around the sun. If the dimensions of the solar system had the same proportions as those of the atom, would the earth be closer to or farther away from the sun than actually it is? The radius of earth's orbit is about 1.5×10^{11} m. The radius of sun is taken as 7×10^{8} m.

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Solution The ratio of the radius of electron's orbit to the radius of nucleus is $(10^{-10}\text{ m})/(10^{-15}\text{ m}) = 10^{5}$, that is, the radius of the electron's orbit is 10^{5} times larger than the radius of nucleus. If the radius of the earth's orbit around the sun were 10^{5} times larger than the radius of the sun, the radius of the earth's orbit would be $10^{5} \times 7 \times 10^{8}$ m = 7×10^{13} m. This is more than 100 times greater than the actual orbital radius of earth. Thus, the earth would be much farther away from the sun.

It implies that an atom contains a much greater fraction of empty space than our solar system does.

Example 12.2 In a Geiger-Marsden experiment, what is the distance of closest approach to the nucleus of a 7.7 MeV α-particle before it comes momentarily to rest and reverses its direction?

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$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{d} = \frac{2Ze^2}{4\pi\varepsilon_0 d}$$

Thus the distance of closest approach d is given by

$$d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$$

The maximum kinetic energy found in α-particles of natural origin is 7.7 MeV or 1.2×10^{-12} J. Since $1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N m^2/C^2. Therefore with $e = 1.6 \times 10^{-19}$ C, we have,

$$d = \frac{(2)(9.0\times10^9\,\mathrm{Nm}^2/C^2)(1.6\times10^{-19}\,C)^2 Z}{1.2\times10^{-12}\,\mathrm{J}}$$

$$= 3.84 \times 10^{-16}\ Z\ \mathrm{m}$$

The atomic number of foil material gold is $Z = 79$, so that

d (Au) = 3.0×10^{-14} m = 30 fm. (1 fm (i.e. fermi) = 10^{-15} m.)

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- Equating forces:

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Kinetic and Potential Energies

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- Potential energy (taking $U(\infty) = 0$):

$$U = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}.$$

ELECTRON ORBITS: TOTAL ENERGY

Result

- Total energy of the electron:

$$E = K + U = \frac{1}{8\pi\epsilon_0}\frac{e^2}{r} - \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}.$$

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- As r increases, $E \to 0^-$ (less tightly bound).

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Solution Total energy of the electron in hydrogen atom is –13.6 eV = $-13.6 \times 1.6 \times 10^{-19}$ J = -2.2×10^{-18} J. Thus from Eq. (12.4), we have

$$E = -\frac{e^2}{8\pi\varepsilon_0 r} = -2.2 \times 10^{-18} \text{ J}$$

This gives the orbital radius

$$r = -\frac{e^2}{8\pi\varepsilon_0 E} = -\frac{(9 \times 10^9 \text{ N m}^2/\text{C}^2)(1.6 \times 10^{-19}\text{C})^2}{(2)(-2.2 \times 10^{-18} \text{ J})}$$

$$= 5.3 \times 10^{-11} \text{ m.}$$

The velocity of the revolving electron can be computed from Eq. (12.3) with $m = 9.1 \times 10^{-31}$ kg,

$$v = \frac{e}{\sqrt{4\pi\varepsilon_0 m r}} = 2.2 \times 10^6 \text{ m/s.}$$